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PROPOSAL DEFENSE

Proxy Validity and Controlled Mitigation in Evidence-Grounded Training for Low-Resource Multilingual Retrieval-Augmented Generation: Evidence from Arabic and Malay Question Answering

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List of Abbreviations

AdaLoRA	Adaptive Low-Rank Adaptation
DoRA	Weight-Decomposed Low-Rank Adaptation
DPO	Direct Preference Optimization
EM	Exact Match
F_{faith}	NLI-based faithfulness score
LoRA	Low-Rank Adaptation
NLI	Natural Language Inference
PEFT	Parameter-Efficient Fine-Tuning
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
QLoRA	Quantized Low-Rank Adaptation
RAG	Retrieval-Augmented Generation
SCST	Self-Critical Sequence Training
VeRA	Vector-based Random Matrix Adaptation

Executive Summary

Problem. Training a Retrieval-Augmented Generation (RAG) model to raise an automatic NLI "is the answer supported?" score can raise the score without improving answers, a form of reward hacking called *proxy divergence*. It is hard to detect in low-resource languages such as Arabic and Malay, where human checking is scarce.

Objectives. O1 (completed): define four RAG failure modes by systematic review (their link to adapters is a hypothesis, not validated). O2 (completed): test an NLI reward with controls and a code audit, on two backbones. O3 (planned): test whether length-controlled DPO can reduce or prevent divergence when it occurs, also under noisy or missing evidence.

Methodology. A PRISMA review of 61 studies with a pilot; a controlled five-seed study (84 runs) of Qwen3-0.6B-Base and T5Gemma 2 after a code audit, with an earlier mT5-base study kept as a baseline; and a staged comparison of six training arms on Qwen3-0.6B-Base, judged by humans and an independent scorer.

Completed. After the audit removed an infrastructure artifact, the reward was inert on both new backbones: its effect on the NLI score was at most 0.012, and nine of twelve intervals included zero, while fine-tuning itself raised F1 and exact match. On mT5-base the same reward raised the Arabic score by $+0.60$ while exact match fell from 0.07 to 0.004; in the small human check, no reliable gain established. An NLI reward can do nothing or do harm, and the score alone cannot tell which. The retrieval pilot is unaudited, and no claim depends on it.

Gap and novelty. Prior work rarely checks whether a grounding reward still tracks quality once optimized, rarely controls length, and has not studied Arabic and Malay. The novelty is narrow: a controlled, audited divergence test and a length-controlled remedy compared with simpler alternatives, not the discovery of reward hacking.

Expected outcomes and significance. A diagnostic protocol, evidence of when divergence occurs, a tested remedy (if it fails, the protocol and evidence remain) and released splits, code and predictions. This helps builders of low-resource RAG avoid trusting an unvalidated score.

3. Introduction

3.1 Research Background

Large pre-trained language models store knowledge in fixed parameters. This creates a persistent tension between parametric memory and factual currency: weights learned during pre-training age and drift, and the models produce fluent but factually incorrect output (Brown et al., 2020; Zhao et al., 2023). Retrieval-Augmented Generation (RAG) was proposed as a direct architectural response. It couples a neural generator to a non-parametric external index queried at inference time (Lewis et al., 2020b), extending earlier retrieval-augmented pre-training such as REALM (Guu et al., 2020) and Atlas (Izacard et al., 2023), and thereby separates knowledge acquisition from model parameters. Within a few years RAG became the dominant pattern for enterprise question answering, legal research, and biomedical decision support.

Wide adoption has nevertheless exposed a systematic failure profile: retrieval does not eliminate hallucination. Magesh et al. (2025) report hallucination rates of 17–33% in commercial legal RAG products, and Niu et al. (2024) annotated hallucinated content in 43.1% of 17,790 responses (29.1% of question-answering responses) from six language models in the RAGTruth corpus, with rates varying strongly by task and model.

Two gaps compound one another and motivate this research. First, the RAG failure literature is overwhelmingly English-centric. Essentially every mitigation strategy covered by the systematic review underlying Study 1 (Section 5.1) is evaluated only on English benchmarks such as Natural Questions (Kwiatkowski et al., 2019), TriviaQA (Joshi et al., 2017), and HotpotQA (Yang et al., 2018). Whether these strategies transfer to typologically distant, lower-resource languages is therefore unknown. Second, within the parameter-efficient fine-tuning (PEFT) literature that this research uses as its intervention mechanism, faithfulness-oriented training rewards are seldom separated from their implementation and evaluation confounds: zero-weight and shuffled-reward controls, reward-independent metrics, and implementation audits are rarely reported. Study 2 of this research illustrates the risk. Its first-draft loss was mathematically non-differentiable, and its first corrected analysis reported an adapter-dependent harm that controls later showed to be an evaluation-state artifact of gradient checkpointing. Both defects would

have silently invalidated the conclusions had they not been detected.

Arabic and Malay were chosen as a typologically diverse evaluation pair. Arabic is Semitic, written right to left, and morphologically rich with trilateral root-and-pattern morphology; Malay is Austronesian, agglutinative, and comparatively under-resourced. Together they stress a mitigation framework across two morphological systems, two scripts, and two resource levels, a combination largely absent from prior RAG evaluations.

The work is organized as a research trajectory of three studies rather than a single experiment.

- **Study 1** (completed; target venue ACM TALLIP) is a PRISMA 2020-conformant systematic review (Page et al., 2021; Moher et al., 2009) of 61 studies. It formalizes four canonical RAG failure modes and proposes a two-stage Low-Rank Adaptation (LoRA) mitigation framework, evaluated in closed-passage pilot experiments on Arabic and a quantitative Malay retrieval baseline (completed; not yet published).
- **Study 2** (completed; prepared for submission to the Journal of King Saud University – Computer and Information Sciences) is a controlled evaluation of an NLI-entailment reward for LoRA fine-tuning of Qwen3-0.6B-Base and T5Gemma 2 270M-270M on Arabic and Malay (five seeds, 84 runs, zero-weight and shuffled-reward controls). The reward changed the NLI score by at most 0.012, and its three nominal exceptions vanished under an independent scorer and at matched length. An earlier mT5-base study is kept as a baseline: there the reward raised the Arabic score by $+0.60$ while exact match fell and answers lengthened without a human-rated gain.
- **Study 3** (planned) is designed to close the limitations that Studies 1–2 could not address within their scope: one reward, a reward that is also the primary metric, no human evaluation of the new backbones, and closed-passage settings in which the answer is guaranteed to be present. Its mandatory design compares six training arms on Qwen3-0.6B-Base, and tests retrieval robustness, with reward-independent evaluation; scaling to larger Qwen3 models is optional.

Figure 1 summarizes how the studies, research questions, and objectives fit together.

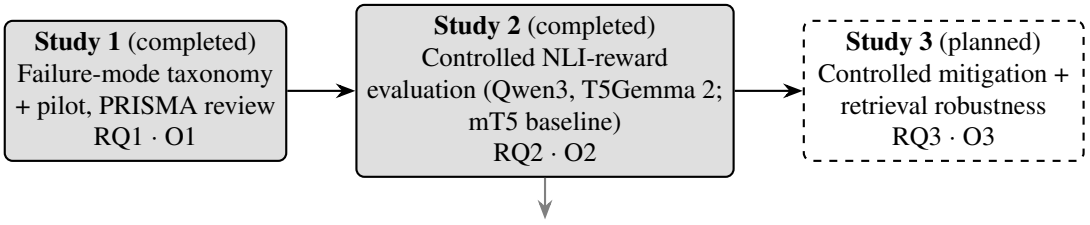


Figure 1: Research trajectory: three linked studies, each answering a research question (RQ) through a specific objective (O).

3.2 Problem Statement

Research issues. Contemporary RAG systems exhibit four technically distinct failure modes that standard cross-entropy training does not explicitly enforce evidence grounding for (Table 1).

Table 1: Four RAG failure modes addressed in this research.

Failure mode	Mechanism
FM1: Missing context	Retrievers trained on English-dominant corpora show a semantic gap for morphologically rich languages, so the correct document is missed entirely.
FM2: Context-window noise	When passages are concatenated, generators lose sensitivity to mid-sequence information, the "lost-in-the-middle" effect (Liu et al., 2024a).
FM3: Formatting errors	Cross-entropy rewards token-level fluency, not structural conformity, producing schema violations in constrained generation.
FM4: Hallucination despite retrieval	Cross-entropy does not explicitly enforce that generated spans are grounded in the retrieved context.

Barnett et al. (2024) independently corroborate a closely related taxonomy from case analysis of production RAG deployments, although without a systematic review protocol or non-English evaluation.

What has been done, and the gap. Standard LoRA (Hu et al., 2022), which builds on a lineage of adapter-based methods (Houlsby et al., 2019; Li and Liang, 2021; Lester et al., 2021; Zaken et al., 2022), addresses parameter efficiency but leaves four limitations: uniform rank allocation across heterogeneous transformer layers, memory demands at scale, coupled magnitude-direction updates that limit format binding, and per-domain adapter overhead that is prohibitive for ultra-low-resource languages. QLoRA (Dettmers et al., 2023), AdaLoRA (Zhang et al., 2023a), DoRA (Liu et al., 2024b), and VeRA (Kopiczko et al., 2024) each target a different subset of these limitations. No prior work, however, maps these variants onto the four failure modes or validates such a mapping

with replicated evidence on a typologically diverse, low-resource language pair.

On the RAG side, mitigation strategies including Fusion-in-Decoder (Izacard and Grave, 2021), FLARE (Jiang et al., 2023), Self-RAG (Asai et al., 2024), CRAG (Yan et al., 2024), Adaptive-RAG (Jeong et al., 2024), and Blended RAG (Sawarkar et al., 2024) each address a different subset of the failure modes, yet the systematic review underlying Study 1 found every one of them evaluated exclusively on English benchmarks. Related work is complementary rather than overlapping: Gao et al. (2023b) offer the most comprehensive RAG taxonomy but neither formalize failure modes mathematically nor evaluate non-English languages; Ding et al. (2023) survey the PEFT landscape without RAG-specific failure contexts; and Es et al. (2024) introduce RAGAs for automated evaluation on English data.

How this work differs, and what is new. Study 1 addresses the mapping and pilot-validation gap. Study 2 addresses the confound gap: it separates the effect of an NLI reward from the engineering that surrounds it, and shows that the NLI score, when it is also the reward, can rise while exact match, answer length and human judgments do not follow. Study 3 is motivated by what Study 2 could not test: when divergence arises (the reward was inert on Qwen3 and T5Gemma 2 but not on mT5-base), whether a preference-based objective can reduce or prevent it when it occurs, and whether either survives imperfect retrieval (distractors, missing evidence and Wikipedia-augmented retrieval), where the answer is not guaranteed to be present. This motivation rests on questions Studies 1–2 left open rather than on a positive result carried forward; Study 3's success is treated as an empirical question, not a presumption.

Antecedent work and choice of methods. This research follows most directly from Lewis et al. (2020b) (the RAG architecture), Hu et al. (2022) and the LoRA-variant lineage (the adaptation mechanism), and Niu et al. (2024) with Ji et al. (2023) (the framing of hallucination despite retrieval). LoRA-family methods were preferred to full fine-tuning and to prompt-based tuning (Lester et al., 2021; Li and Liang, 2021) because they fit the single-GPU, small-fine-tuning-set regime that motivates this work, consistent with the intrinsic-dimensionality argument of Aghajanyan et al. (2021); Study 2 found no evidence that the choice among LoRA, QLoRA and DoRA changes how a reward acts.

Direct Preference Optimization (Rafailov et al., 2023) is selected for Study 3 because the scalar NLI reward of Study 2 was optimized into a regime where the NLI score rose (on both the training and an independent NLI scorer) while exact match and blinded human judgments did not show a corresponding improvement, whereas DPO recasts the grounding objective as a pairwise preference loss that removes the manually tuned penalty weight (λ_1) and the sampled-reward estimator, provided the preference pairs are controlled for answer length.

Central question and principal contribution. The thesis asks one question: *when does an evidence-based training signal improve a grounding score without improving the quality of the answer, and can that mismatch be detected and reduced?* The motivating evidence comes from Study 2: on the legacy mT5-base baseline, an NLI-entailment reward raised the Arabic NLI score by $+0.60$ while exact match, answer length and human ratings did not follow, whereas on Qwen3-0.6B-Base and T5Gemma 2 the same reward had almost no effect (at most 0.012). The **principal contribution** is a *diagnostic protocol* that detects this proxy divergence: separate measures of support, correctness and length, controls, human rating and a falsifiable rule for when divergence is concluded (Section 5.3). The protocol has been applied to three backbones (human evidence for mT5-base only, one reward setting), and Study 3 tests when divergence occurs on Qwen3-0.6B-Base and whether the proposed remedy, length-controlled DPO, can reduce or prevent it. If the remedy fails, or a simple length control matches it, the protocol and the divergence evidence remain the thesis result. Table 3 separates what is demonstrated, suggestive, proposed and untested, and Section 4.9 states how Study 3 differs from the closest prior work.

Constructs and primary outcomes. The word "faithfulness" is used in this thesis only for the NLI score F_{faith}, which is a proxy. Answer quality is decomposed into the separate constructs of Table 2, each with its own measure, so that an improvement in one is never reported as an improvement in another.

Table 2: Constructs, measures and primary outcomes.

Construct	Definition	Measure	Primary in
Evidence support	Answer content is entailed by the supplied passage	NLI score (proxy); human support rating (0–2)	Study 2 (proxy F_{faith}, as reward)
Question relevance	Answer addresses the question asked	Human rating; question–answer match	Study 3 (component)
Correctness	Answer matches the gold answer	Exact match, token-F1; human rating	Study 3 (non-inferiority)
Completeness	Answer states what the question requires, no more or less	Human rating (answer length is a diagnostic only)	Study 3 (length bound)
Abstention	System declines when evidence is insufficient	Correct-abstention rate under missing evidence	Study 3 (robustness)

Study 3 primary outcome: human-judged "supported-and-correct" rate

Table 3: Status of the principal claims at the time of the proposal defense.

Claim	Status	Evidence	Main limit
The NLI reward adds no detectable NLI gain on Qwen3-0.6B-Base or T5Gemma 2	Demonstrated (two small backbones)	Effect ≤ 0.012; 9 of 12 intervals include 0; gains above about 0.03 excluded	$\lambda_1 = 0.3$, small models, one split
The reward is inert because cross-entropy training memorizes 470 examples	Hypothesis from logs	Policy-gradient term below 10^{-3} in 80–100% of late steps	Untested; needs within-prompt reward variance
Fine-tuning, not the reward, lowers NLI through longer answers	Demonstrated (descriptive)	Matched-length contrasts ≈ 0; F1 and EM gains large	Length associated, not proved causal
On mT5-base the reward raised the Arabic score while EM fell	Demonstrated (legacy baseline)	$+0.60$ (CI $+0.56$ to $+0.64$); EM 0.07 to 0.004; 2.9 to 16.5 words; controls within 0.01	Near-floor baseline; one backbone
Human-rated quality did not improve (mT5)	Suggestive only	60 items per language, $\kappa = 0.38$, mean support difference -0.22	Small, low agreement; none for new backbones
Arabic and Malay differ because of language	Not established	Arabic (extractive) and Malay (multiple-choice-derived) differ in task	Language and benchmark confounded
Failure-mode taxonomy	FM4 tested; FM1 and FM3 partly; FM2 untested	Table 7	Mapping of LoRA variants to modes is a hypothesis
The protocol detects divergence in general	Proposed (Study 3, test D)	None yet beyond Study 2	One backbone so far
Length-controlled DPO is a remedy	Proposed (Study 3)	None	Untested

3.2.1 Research Questions

RQ1. How can the failure modes of Retrieval-Augmented Generation be formally characterized as identifiable breakdowns in the retrieval-generation factorization, and which Low-Rank Adaptation variants are theoretically best suited to each? *(Study 1)*

RQ2. Once implementation and evaluation confounds are controlled, when does an NLI-entailment training reward improve the evidence-support score without improving correctness, answer form or human-judged quality, and does the LoRA variant modulate this? *(Study 2)*

RQ3. Does a length-controlled preference objective, compared against cross-entropy training, a scalar NLI reward, ordinary DPO and a simple length-control baseline on the same backbone and data, improve human-judged supported-and-correct answers without correctness loss or length inflation, and do its benefits survive missing, distracting and actually retrieved evidence? *(Study 3)*

3.2.2 Research Hypotheses

H1 (Studies 1–2). Training with an NLI-entailment reward under LoRA fine-tuning improves measured faithfulness (F_{faith}, an NLI score) and also reference-based correctness (exact match, token-F1) and human-judged support. *Outcome:* not supported. On Qwen3-0.6B-Base and T5Gemma 2 the reward changed the score by at most 0.012; on mT5-base it raised the Arabic score without improving correctness or support. *Independent variables:* training configuration (cross-entropy LoRA, composite NLI-reward LoRA, with zero-weight and shuffled-reward controls), backbone (Qwen3-0.6B-Base, T5Gemma 2; mT5-base as baseline), and language (Arabic, Malay). *Dependent variables:* F_{faith}, exact match, token-F1, answer length, and blinded human ratings of support and correctness. H1 was stated as a joint claim so that a gain in the proxy alone could not be mistaken for a gain in quality.

Outcome. H1 was **not supported**. On Qwen3-0.6B-Base and T5Gemma 2 the reward changed the NLI score by at most 0.012 (nine of twelve intervals include zero; the three Qwen3 Arabic exceptions vanish under an independent scorer and at matched answer length), while fine-tuning itself raised F1 and exact match. On the legacy mT5-base baseline the reward raised the Arabic NLI score by $+0.60$ (95% CI $+0.56$ to $+0.64$), specifically because of the reward (controls within 0.01), but exact match fell from 0.07 to 0.004, answers grew from 2.9 to 16.5 words, and blinded raters judged them no more correct and, on average, less supported; the Malay change was $+0.008$. These outcomes show that the score alone cannot separate an inert reward from a harmful one, which

motivates H2.

H2 (Study 3). On Qwen3, a length-controlled DPO objective improves independently measured answer quality relative to supervised fine-tuning, and its advantage is not explained by answer length. H2 is split into five separately decided parts, with margins fixed before any run (Table 10): **H2a** the human-judged supported-and-correct rate improves; **H2b** token-F1 (the primary correctness metric) is non-inferior, with exact match as a secondary check (H2b′); **H2c** mean answer length stays within a stated bound; **H2d** support and F1 remain non-inferior under distractors and actual retrieval; **H2e** abstention is correct when the supplied evidence is missing, without excess false abstention. H2d and H2e are secondary and descriptive. A divergence test D applies the exclusive rules of Table 14 to the NLI-reward arm against SFT, using the same double-rated primary block. A control hypothesis compares length-controlled DPO with the length-control baseline and with a volume-matched ordinary DPO, so that a gain attributable to length or training volume alone is detected; failure to establish an advantage over a control is reported as no demonstrated additional benefit, not as evidence of equivalence. *Independent variables:* training objective (SFT, scalar NLI reward, ordinary DPO, volume-matched ordinary DPO, length-controlled DPO, length-control baseline) and evidence condition (oracle, distractors, missing, retrieved); backbone scale (Qwen3 0.6B, with 1.7B and 8B as optional confirmation). *Dependent variables:* supported-and-correct rate (human), exact match, F1, answer length, correct-abstention rate, retrieval recall.

3.3 Research Aim and Objectives

The aim of this research is to establish when evidence-grounding training signals diverge from independently measured answer quality in low-resource, typologically diverse languages, and to test a controlled mitigation. The work proceeds from a review-grounded pilot, through a controlled, audited closed-passage evaluation, to a staged, controlled mitigation and retrieval-robustness study on Qwen3-0.6B-Base. The objectives are stated against SMART criteria (Specific, Measurable, Achievable, Relevant, Time-bound) and mapped to the research questions in Table 4.

O1. **[Completed: Study 1]** Conduct a PRISMA 2020-conformant systematic review of RAG failure modes and PEFT mitigation strategies across six databases (2019–2025); formalize four failure modes as measurable breakdowns in the retrieval-generation factorization; and test a two-stage QLoRA-based mitigation framework in closed-passage pilot experiments on Arabic (XQuAD) and a Malay (Belebele) retrieval baseline (only the parts recorded as tested in Table 7 have been evaluated). *Completed in Semester 1. Addresses RQ1.*

O2. **[Completed: Study 2]** Evaluate an NLI-entailment reward for LoRA fine-tuning of Qwen3-0.6B-Base and T5Gemma 2 270M-270M on the Arabic and Malay closed-passage test sets, with five seeds, zero-weight and shuffled-reward controls, a matched-length analysis and an independent scorer, after auditing and correcting the training and evaluation code; mT5-base results are retained as a baseline. *Completed by Semester 2. Addresses RQ2.*

O3. **[Planned: Study 3]** In four stages: (3a) re-run the corrected protocol on Qwen3-0.6B-Base; (3b) compare six arms (SFT, a scalar NLI reward, ordinary DPO, a volume-matched ordinary DPO, length-controlled DPO and a length-control baseline) on that backbone, data and candidate pool; (3c) hold the trained generator fixed and test oracle, distractor, missing and retrieved evidence; and (3d, optional) extend the strongest comparison to Qwen3-1.7B and 8B. *Targeted for Semesters 3–5. Addresses RQ3.*

Table 4: Alignment of research questions, objectives, studies, methods, and status.

RQ	Objective	Study	Hypothesis	Primary method	Status
RQ1	O1	Study 1	(framework)	PRISMA review; QLoRA pilot	Completed
RQ2	O2	Study 2	H1	5-seed, 84-run controlled study with audit	Completed
RQ3	O3	Study 3	H2	Staged controlled comparison; retrieval robustness	Planned

3.4 Research Scope

The research is scoped to text-based question answering in two languages: Modern Standard Arabic (evaluated on XQuAD) and Standard Malay in Latin script (evaluated on Belebele). Studies 1–2 operate under a *closed-passage* assumption, in which the

correct answer is guaranteed to lie within a bounded passage pool (a pool of 1,190 context entries for Arabic XQuAD, one per question, covering 240 unique paragraphs; the effect of this duplication on retrieval metrics is reported with the Study 1 results). This yields an optimistic upper bound on retrieval performance rather than a realistic open-domain estimate, and is the primary motivation for the retrieval-robustness stage of Study 3.

The generator backbones are Qwen3-0.6B-Base and T5Gemma 2 270M-270M in Study 2 (mT5-base, 580M, is the baseline from the earlier study), and Qwen3-0.6B-Base in Study 3 (Qwen3-1.7B and 8B are optional extensions), chosen because Qwen3 is pre-trained on about 119 languages, including Arabic and Malay. The adaptation method is LoRA (rank 8) in Study 2 (LoRA variants were compared only in the mT5 baseline), and QLoRA, with DoRA as a secondary check, in Study 3. Faithfulness is operationalized throughout as an NLI-entailment metric (F_{faith}) rather than human-annotated factuality, and the known susceptibility of such proxies to gaming (Section 4.5) is treated as an explicit scope boundary. Dialectal Arabic, languages beyond Arabic and Malay, and multi-hop or conversational RAG are out of scope and noted as future work.

One further limitation concerns benchmark comparability. Belebele is a four-way multiple-choice reading-comprehension benchmark, whereas XQuAD is extractive question answering. This task-format asymmetry is a confound in any cross-language comparison, and acknowledging it does not remove it. Study 3 therefore adds a harmonized arm, Belebele in both Arabic (arb_Arab) and Malay (zsm_Latn) with identical format and prompt, on which alone cross-language contrasts are drawn; XQuAD-Arabic is reported as a separate extractive arm, and all other conclusions are restricted to each language–benchmark pair. The confound affects the Arabic-versus-Malay contrast observed in Study 2 (Section 5.2) and is a stated limitation, not an assumption of equivalence.

3.5 Potential Contribution

The principal contribution is a controlled, audited test of when an NLI grounding reward diverges from independently judged answer quality in Arabic and Malay, together with a diagnostic protocol (pre-specified decision rules, controls and a blinded human check) and a length-controlled preference remedy compared against simpler alternatives. The

completed work adds a transparent record of how two implementation defects first produced a misleading conclusion that was then corrected, which sets a standard of evidence uncommon in the PEFT literature and speaks to reproducibility concerns across machine learning (Ioannidis, 2005). If the remedy fails, the protocol and the divergence evidence remain the result.

The supporting contributions are a PRISMA-grounded taxonomy of four RAG failure modes with pilot evidence on a non-English, low-resource language pair (only the modes flagged as tested in Table 7 have been evaluated), and a Malay passage-retrieval baseline on a Belebele-derived pool, for which no earlier report was found (a resource contribution on which no scientific claim depends; consistent with benchmark coverage such as MIRACL (Zhang et al., 2023b), which spans 18 languages but not Malay). Study 3 also reports latency, memory and hosting cost, so that adoptability can be judged, and aims to support more equitable information access for speakers of languages under-served by current NLP tooling.

4. Literature Review

The review is organized around seven themes that together form the theoretical framework of this research: (1) transformer and pre-trained language model architectures; (2) RAG architectures and their evaluation; (3) RAG failure modes and mitigation; (4) the PEFT and LoRA family; (5) hallucination and faithfulness evaluation; (6) low-resource and multilingual NLP, with emphasis on Arabic and Malay; and (7) preference optimization and decoder-only large language models, which ground the planned Study 3. Within each theme, prior work is assessed critically: what has been done, where it falls short, and which gap the three studies address.

4.1 Foundational Architectures

The Transformer (Vaswani et al., 2017) underlies essentially all work reviewed here. Encoder-only pre-training (Devlin et al., 2019) and its multilingual extensions (Conneau et al., 2020; Lample and Conneau, 2019) produced the masked-language-model representations widely used for retrieval and classification. Encoder-decoder pre-training (Raffel et al., 2020) and its multilingual variant mT5 (Xue et al., 2021) established the

sequence-to-sequence formulation used as the generator in Studies 1–2. Decoder-only, autoregressive pre-training at scale (Brown et al., 2020), and its instruction-tuned and multilingual descendants (Touvron et al., 2023; Dubey et al., 2024; Aryabumi et al., 2024; Gemini Team, 2024; OpenAI, 2023), define the architecture family targeted by Study 3; Zhao et al. (2023) survey this trajectory. Across all of these architectures, parametric knowledge is fixed at pre-training time and cannot be cheaply updated, corrected, or audited. This is the limitation that RAG was designed to remove.

4.2 Retrieval-Augmented Generation: Architectures and Evaluation

Lewis et al. (2020b) formulated RAG as a marginalization over a retrieved document distribution,

$$p(y \mid x) = \sum_{z} p_{\eta}(z \mid x)\, p_{\theta}(y \mid x, z), \tag{1}$$

building on retrieval-augmented pre-training: REALM (Guu et al., 2020) integrates retrieval into masked-language-model pre-training, and Atlas (Izacard et al., 2023) extends the approach to few-shot settings. Dense Passage Retrieval (Karpukhin et al., 2020) established the dominant bi-encoder paradigm, subsequently scaled with efficient similarity search (Johnson et al., 2021) and assessed through benchmarks such as MTEB (Muennighoff et al., 2023) and sentence-embedding methods (Reimers and Gurevych, 2019). Fusion-in-Decoder (Izacard and Grave, 2021) and Retro (Borgeaud et al., 2022) explored architectures for combining multiple retrieved passages, and earlier open-domain question answering (Chen et al., 2017) established the retrieve-then-read pattern that RAG generalizes.

More recent adaptive strategies (Self-RAG (Asai et al., 2024), FLARE (Jiang et al., 2023), CRAG (Yan et al., 2024), Adaptive-RAG (Jeong et al., 2024), Blended RAG (Sawarkar et al., 2024), and REPLUG (Shi et al., 2023b)) change when or how retrieval is invoked. The systematic review underlying Study 1 found that every one of these methods is evaluated exclusively on English benchmarks (Natural Questions (Kwiatkowski et al., 2019), TriviaQA (Joshi et al., 2017), HotpotQA (Yang et al., 2018), MS MARCO (Bajaj et al., 2016)). Surveys (Gao et al., 2023b; Zhu et al., 2023; Wang et al., 2024) consolidate the architectural landscape but do not formalize failure modes or address multilingual evaluation, which is the gap Study 1 targets. Automated frameworks

such as RAGAs (Es et al., 2024) assess answer relevance and groundedness, but were developed and validated on English data only.

4.3 RAG Failure Modes and Mitigation

That retrieval does not eliminate hallucination is now well documented. Niu et al. (2024) annotated hallucination in 43.1% of 17,790 RAGTruth responses overall (29.1% for question answering) across six models, with rates varying by task and model, and Magesh et al. (2025) independently found hallucination rates of 17–33% in production legal-RAG products. Barnett et al. (2024) identify seven engineering failure points from three production deployments, giving convergent practitioner-level evidence for a failure taxonomy, though without a systematic protocol, formalization, or non-English evaluation. Shi et al. (2023a) and Liu et al. (2024a) show that generators are measurably distracted by irrelevant or mispositioned context, which directly evidences the context-window-noise mode. Mallen et al. (2023) show that retrieval helps least when parametric memory is already reliable, which motivates adaptive retrieval.

Study 1 synthesizes this literature into a four-mode taxonomy (missing context, context-window noise, formatting errors, and hallucination despite retrieval), each formalized with a distinct measurable metric (Section 5.1). It then maps four LoRA variants to the failure mode each is theoretically best placed to address. QLoRA's memory efficiency allows retriever and reranker adaptation on consumer hardware, and AdaLoRA's adaptive rank allocation concentrates capacity in retrieval-relevant layers (FM1). DoRA's magnitude-direction decomposition improves format-conformity binding (FM3), and VeRA's shared projections suit ultra-low-resource adaptation. No prior systematic review identified in this literature search performs this mapping with empirical validation on a typologically diverse, low-resource language pair.

4.4 Parameter-Efficient Fine-Tuning and the LoRA Family

PEFT methods adapt large pre-trained models by updating only a small fraction of their parameters. Adapter methods (Houlsby et al., 2019) insert bottleneck layers between transformer blocks; prefix and prompt tuning (Li and Liang, 2021; Lester et al., 2021) learn continuous input representations; BitFit (Zaken et al., 2022) updates only bias

terms; and IA3 (Liu et al., 2022) rescales activations with learned vectors. He et al. (2022) offer a unifying framework across these methods, and Ding et al. (2023) and Han et al. (2024) survey the wider landscape without addressing RAG-specific failure contexts.

Low-Rank Adaptation (Hu et al., 2022) reparameterizes the weight update as a low-rank product $\Delta W = BA$, motivated by the low intrinsic dimensionality of fine-tuning updates (Aghajanyan et al., 2021). The family has since diversified:

- QLoRA (Dettmers et al., 2023) combines LoRA with 4-bit backbone quantization, enabling fine-tuning of large models on consumer hardware with little accuracy loss;
- AdaLoRA (Zhang et al., 2023a) casts the update in SVD form and reallocates rank budget across layers by singular-value importance;
- DoRA (Liu et al., 2024b) decomposes the update into magnitude and direction components, improving fine-tuning quality without added inference cost;
- VeRA (Kopiczko et al., 2024) shares frozen random projections across layers, cutting trainable parameters roughly tenfold relative to standard LoRA;
- PiSSA (Meng et al., 2024) initializes the low-rank factors from the principal singular directions of the pre-trained weights; and
- rank-stabilized LoRA (Kalajdzievski, 2023) corrects a scaling instability at high rank.

Each variant addresses a different subset of the limitations left open by standard LoRA (uniform rank allocation, memory footprint, coupled magnitude-direction updates, and per-domain adapter overhead). Before Study 1, however, no work had mapped this family onto a RAG failure-mode taxonomy or tested such a mapping under statistically replicated conditions.

4.5 Hallucination and Faithfulness Evaluation

Faithfulness evaluation spans several methodological families. Surface-overlap metrics such as BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004) measure lexical similarity to a reference but correlate poorly with factual grounding, and embedding-based metrics such as BERTScore (Zhang et al., 2020) still do not test entailment. Natural Language

Inference (NLI)-based metrics, adopted throughout Studies 1–2, test whether a generated span is entailed by its source context. Maynez et al. (2020) established this framing for abstractive summarization, Laban et al. (2022) extended it with SummaC, and Ji et al. (2023) and Huang et al. (2023) systematize it in surveys of hallucination. QA-based consistency metrics (Fabbri et al., 2022) and atomic fact scoring (Min et al., 2023) provide complementary, costlier protocols. The operational metric F_{faith} rests on cross-lingual and multi-genre NLI resources (Conneau et al., 2018; Williams et al., 2018), an efficient classifier (He et al., 2021), and transfer-learning-based deployment (Laurer et al., 2024).

Any span-overlap or entailment proxy carries a recognized limitation: it can be gamed by degenerate generation. Study 1's own "H_{proxy} paradox" finding illustrates this, since a frozen baseline that copies long extractive spans verbatim mechanically minimizes the hallucination proxy without producing a correct answer. This research therefore treats the limitation as an explicit scope boundary (Section 3.4).

4.6 Low-Resource and Multilingual NLP: Arabic and Malay

Arabic NLP has comparatively mature infrastructure. AraBERT (Antoun et al., 2020) and the CAMeLBERT family (Inoue et al., 2021) provide strong Arabic-specific representations, and Habash (2010) remains a standard reference for Arabic computational morphology. Relevant benchmarks include XQuAD (Artetxe et al., 2020), the primary Arabic closed-passage evaluation set in Studies 1–2; MLQA (Lewis et al., 2020a); TyDi QA (Clark et al., 2020); XOR QA (Asai et al., 2021); and the Arabic-specific ARCD (Mozannar et al., 2019).

Malay resources are comparatively scarce. Belebele (Bandarkar et al., 2024), built from FLORES-200 (NLLB Team et al., 2022), serves as the Malay evaluation set because the systematic review found no native extractive Malay QA benchmark comparable to XQuAD. The closest widely used multilingual retrieval benchmark, MIRACL (Zhang et al., 2023b), covers 18 languages but not Malay, and SEACrowd (Lovenia et al., 2024) addresses Southeast Asian language resources only in general terms, not with an extractive-QA benchmark. Broader treatments of low-resource NLP (Magueresse et al., 2020) indicate that this asymmetry is structural rather than incidental. It supports the deliberate choice of a typologically diverse, comparatively low-resource pair over a

higher-resource comparison such as French or Spanish.

4.7 Preference Optimization and Decoder-Only LLMs for RAG

Study 3 rests on two literatures largely separate from those underpinning Studies 1–2. The first concerns decoder-only multilingual and instruction-tuned models: Qwen3 (Qwen Team, 2025) (pre-trained on about 119 languages, including Arabic and Malay), Aya (Üstün et al., 2024), Aya-23 (Aryabumi et al., 2024), Llama 2 (Touvron et al., 2023), the Llama 3 herd (Dubey et al., 2024), and proprietary systems such as GPT-4 (OpenAI, 2023) and Gemini 1.5 (Gemini Team, 2024). These now form the dominant family for practical RAG deployment, in contrast to the encoder-decoder mT5-base used in the earlier Study 2 baseline; Study 3 builds on the Study 2 results for Qwen3-0.6B-Base (mandatory) and, if resources allow, T5Gemma 2 (Zhang et al., 2025), a recent encoder-decoder (optional).

The second is preference-based fine-tuning. Direct Preference Optimization (Rafailov et al., 2023) recasts alignment as a closed-form pairwise preference loss derived from the reward-modeling objective of reinforcement learning from human feedback, removing the need for an explicit reward model or on-policy sampling. This bears directly on limitations met in Study 2: the composite objective required a manually specified penalty weight (λ_1), and its leave-one-out REINFORCE estimator (Rennie et al., 2017; Ahmadian et al., 2024) optimized a scalar NLI reward into a regime where the score rose while exact match fell and answers grew about sixfold, the pattern described as reward over-optimization in preference learning (Gao et al., 2023a). A preference-based formulation removes the sampled reward and the penalty weight, but it keeps its own trade-off parameter β (the strength of the implicit KL constraint to the reference model) and is not immune to length bias (Park et al., 2024), so Study 3 treats β as a tuned hyperparameter and controls preference pairs for answer length. More recent preference-optimization variants (for example ORPO, KTO, and SimPO) also address hyperparameter sensitivity and are candidate comparators if resources permit; DPO is retained as the primary method because it is the best established and most widely replicated.

Adaptation in Study 3 uses QLoRA (Dettmers et al., 2023), which makes the

8-billion-parameter Qwen3 model trainable on a single GPU, with DoRA (Liu et al., 2024b) as a secondary check. A combined quantized, weight-decomposed strategy is no longer proposed as a contribution, because Study 2 found no evidence that the choice among LoRA, QLoRA and DoRA changes how a reward acts.

4.8 Systematic Review Methodology

The review underlying Study 1 follows PRISMA 2020 (Page et al., 2021), the current reporting standard, which updates the original 2009 statement (Moher et al., 2009), and adapts established software-engineering review procedures (Kitchenham, 2004) to computational linguistics and information retrieval. A structured Boolean search of six databases (ACL Anthology, IEEE Xplore, Scopus, Google Scholar, Semantic Scholar, and arXiv cs.CL/cs.IR; January 2019 to April 2025) reduced 847 initial records to 61 included studies through de-duplication, title and abstract screening, and full-text assessment, using explicit, pre-registered inclusion and exclusion criteria. The protocol and its corpus underpin the failure-mode taxonomy and LoRA-variant mapping summarized above and detailed in Section 5.1.

4.9 Closest Related Work and Positioning

This subsection summarizes, in plain terms, how the closest prior work relates to Study 3; the detailed comparison and the dated evidence record are in Appendix C. The systematic review of Study 1 closed in April 2025, so a targeted update search to 17 October 2026 was added. It is a documented targeted search, not a replication of the PRISMA protocol, and the original-database re-run is still planned before submission.

Novelty in one sentence. As far as the search to 17 October 2026 shows, no identified study combines an entailment-based training signal, a length-controlled preference remedy compared with ordinary, volume-matched and length-baseline controls, and human rating in Arabic and Malay. The thesis does not claim to be the first to observe reward hacking; its claim is a controlled, audited test of it in these languages and a remedy that is checked against simple alternatives.

Table 5: Literature gaps at a glance (prior work, what it lacks, and what this thesis adds).

Prior work	What it did	What it lacks	Our addition
Checker and faithfulness rewards for RAG (Ji et al., 2026; Xu et al., 2025; Pham et al., 2026; Wang et al., 2026; Zhao et al., 2026)	Train with checker, rubric or gated rewards; reward hacking or answer-less-instead trade-offs observed in English	No shuffled-reward control, code audit or human check in Arabic and Malay	Audited, controlled test with a human check (Study 2)
Preference training for robust RAG (Yan et al., 2025; Zhang et al., 2024; Kachuee et al., 2025; Huang et al., 2025)	DPO or supervised variants for noisy or distracting context, mostly English	No length-matched pairs and no volume-matched or length-baseline comparison	Length-controlled DPO against ordinary, volume-matched and length-baseline controls (Study 3)
Length bias in RLHF and DPO (Singhal et al., 2023; Chen et al., 2024b; Park et al., 2024; Lu et al., 2024; Li et al., 2025)	Length exploitation documented and debiased in general alignment (reward hacking is formalized by Skalse et al. 2022)	Not tested on evidence-grounded RAG or on Arabic and Malay	Length control built into pair construction for grounded answers
Multilingual and Arabic RAG evaluation	Cross-lingual retrieval and answering; abstention in English	No study of training signals; Malay largely absent	Arabic and Malay under one protocol

4.10 Summary of Gaps

Table 6 consolidates how the reviewed literature leads to this research.

Table 6: Dated, scoped gaps and the study that addresses each (statements are bounded by the searches described in Section 4.9).

What has been done	What has not been done	Addressed by
RAG architectures and mitigations (Self-RAG, CRAG, FLARE), as covered by the Study 1 review (to April 2025)	Within that corpus, no evaluation beyond English and no formal failure-mode taxonomy were found	Study 1
LoRA-family methods (QLoRA, AdaLoRA, DoRA, VeRA), as covered by the review	Within that corpus, no mapping to RAG failure modes and no controlled low-resource replication; the mapping remains a hypothesis (Table 7)	Studies 1–2
NLI-based faithfulness metrics and checker rewards	Reward hacking with checker rewards is documented in English biomedical RAG (Ji et al., 2026); as far as found to 17 October 2026, not with a shuffled-reward control, an implementation audit and a blinded human check in Arabic and Malay (a zero-weight control appears in that work)	Study 2
Preference training for grounded RAG (RAFT, RPO and later work in Table 20)	As far as found to 17 October 2026, no length-controlled preference construction tested against ordinary, volume-matched DPO and a length baseline, with human rating, for Arabic and Malay	Study 3

5. Research Methodology

The research combines three methodologically linked designs that correspond to Objectives O1–O3 (Section 3.3): a systematic literature review with embedded pilot experimentation (Study 1), a controlled, audited multi-seed experiment (Study 2), and a planned staged comparison of preference-based and reward-based fine-tuning with retrieval-robustness tests (Study 3). For each, this section states the methodology, justifies the choice of methods against alternatives, and records the status of the work.

5.1 Objective O1 (Completed): Systematic Review and Pilot Validation

5.1.1 Methodology

Study 1 followed a PRISMA 2020-conformant protocol (Page et al., 2021). A structured Boolean search combined three query dimensions: the RAG paradigm; the failure taxonomy (hallucination, missing context, context noise, formatting error); and the adaptation mechanism (LoRA, QLoRA, AdaLoRA, DoRA, PEFT). It was run across six databases (ACL Anthology, IEEE Xplore, Scopus, Google Scholar, Semantic Scholar, and arXiv cs.CL/cs.IR) for January 2019 to April 2025. Screening proceeded through four stages (search, de-duplication, title and abstract screening, full-text assessment) and reduced 847 initial records to 61 included studies. The 128 exclusions at the full-text stage were recorded by category: non-methodological application papers, non-peer-reviewed grey literature, papers lacking quantitative evaluation, papers outside the temporal window, and miscellaneous.

The corpus was then used to formalize the four failure modes mathematically (Section 4.3) and to design a two-stage mitigation framework. Stage 1 (retrieval adaptation, targeting FM1–FM2) fine-tunes a hybrid BM25 and dense retriever and an mT5-base reranker with a contrastive objective. Stage 2 (generation adaptation, targeting FM3–FM4) fine-tunes the mT5-base generator with a composite loss that combines cross-entropy with a grounding penalty. Pilot closed-passage experiments evaluated Stage 1 on Arabic XQuAD (a pool of 1,190 context entries, one per question, covering 240 unique paragraphs) and established a BM25 baseline on Malay Belebele. Only FM1 and, in Study 2, FM4 were tested empirically; Table 7 states the status of each mode.

Table 7: Operational definitions and empirical status of the four failure modes.

Mode	Identification rule	Metric	Status
FM1	Gold evidence absent from the top-k retrieved passages	Recall@k, hit rate	Tested in the closed pool (Arabic XQuAD, Malay BM25); not tested at open-domain scale (Study 3c)
FM2	Accuracy falls when distractors are added to the gold passage or the gold position changes	ΔF1 against oracle-only context, by distractor count and position	Defined; not yet tested (Study 3c)
FM3	Output violates the required form (not an answer span or option text; sentinel tokens; inflated length)	Format-violation rate; answer length against gold	Observed incidentally in Study 2 (sentinel suppression, length inflation); schema-constrained generation untested
FM4	Answer content not entailed by the supplied passage	Human support rating; NLI score as proxy	Tested in Study 2 on closed passages; human check small

The generation-side runs of the first protocol (mT5-base, 17 August to 12 October 2026) are superseded by the code audit described under Study 2 and are not reported as results. The Stage 1 retrieval pilot (QLoRA adaptation of the retrieval stage, described above) ran in a separate pipeline that has not been independently audited and whose code is not in the audited Study 2 repository; its result (Arabic XQuAD retrieval F1 from 0.050 to 0.187) is therefore preliminary, no thesis claim depends on it, and a component-level record is being prepared as Supplementary S2. Appendix A gives the dated chronology.

The Study 1 mapping of LoRA variants to failure modes is a theoretical hypothesis. Study 2 found no adapter dependence for FM4, and the mappings for FM1–FM3 remain untested; the four modes are an *operational taxonomy* used to organize the experiments, not a validated or exhaustive classification, and nothing is claimed beyond what Table 7 records.

5.1.2 Justification

A systematic review, rather than a narrative survey, was chosen because the research questions (which failure modes exist, which are already mitigated, and where the evidence is missing) require a reproducible and auditable search-and-screening protocol. PRISMA 2020 was selected because it is the current, widely endorsed reporting standard that supersedes the 2009 statement (Moher et al., 2009), and its explicit screening stages produce the inclusion and exclusion counts needed to substantiate a claimed gap rather than assert one impressionistically. QLoRA was chosen as the pilot's adaptation method

because 4-bit quantization made fine-tuning feasible on the available consumer-grade and free-tier cloud GPUs (Google Colab, NVIDIA T4 and A100), a constraint that is itself characteristic of low-resource-language research.

5.2 Objective O2 (Completed): Controlled Evaluation of an NLI Reward for LoRA Fine-Tuning

5.2.1 Methodology

Study 2 extended the Stage 2 pilot of Study 1 into a controlled closed-passage evaluation. Cross-entropy training (configuration B) is compared with a composite objective (configuration C) that adds a REINFORCE-style term rewarding the NLI-entailment score F_{faith} of the model's own sampled answers ($\lambda_1 = 0.3$, $K = 4$ samples; a frozen NLI model supplies a scalar reward only). The exact estimator and training settings are given in Appendix B.

Code audit. A first study (four LoRA variants, three seeds, 188 runs) appeared to show an adapter-dependent collapse under the reward. Zero-weight and shuffled-reward controls and a gradient-checkpointing check showed that the collapse was an infrastructure artifact (a T5 model left in training mode with checkpointing enabled decodes from the last token only). Six defects were corrected, each with a test that fails on the old behavior, and the first study is kept as exploratory only.

Primary study. The primary study uses Qwen3-0.6B-Base and T5Gemma 2 270M-270M on Arabic XQuAD and Malay Belebele, LoRA (rank 8, $\alpha = 16$), five seeds (42, 123, 2026, 7, 99), a split of 470 training, 720 Arabic and 430 Malay held-out examples, and five conditions: frozen (A), cross-entropy (B), composite (C), zero-weight ($C_{\lambda=0}$) and shuffled-reward (C_{shuf}) (Table 8). The primary contrast is the seed-paired change in F_{faith} from B to C; exact match, F1, answer length, an independent NLI scorer and a length-matched analysis are the reward-independent outcomes. No human evaluation was run on the new backbones.

Findings. (i) The reward had little effect on the NLI score: at most 0.012 on either backbone, nine of twelve paired contrasts include zero, and the three exceptions (all Qwen3, Arabic: $+0.011$, $+0.012$, $+0.008$) were not reproduced by an independent scorer

Table 8: Study 2 design: primary study and legacy mT5-base baseline.

Component	Content	Runs
Primary study	2 backbones × 2 languages × (4 trained conditions × 5 seeds + 1 frozen)	**84**
Legacy mT5-base baseline	LoRA, QLoRA, DoRA × 2 languages × 8 seeds, with controls and checkpointing check	162 jobs, 228 records
First study (exploratory)	Four variants, 3 seeds	188

(+0.002, 95% CI −0.004 to +0.009) and vanish at matched answer length (+0.003, CI −0.004 to +0.009). Noise reaches 0.012 in NLI, so gains above about 0.03 are excluded. (ii) Fine-tuning itself gives large F1 and exact-match gains, and most of the NLI drop after fine-tuning is explained by answer length. (iii) The policy-gradient term was below 10^{-3} in 80–100% of second-half logged steps. A hypothesis from the logs, not tested, is that cross-entropy training memorizes the 470 examples, so sampled answers carry no reward variance; a reward needs a stochastic policy, and logging within-prompt reward variance is the recommended diagnostic. (iv) On the legacy mT5-base baseline, which does not memorize, the reward raised the Arabic score by +0.60 (95% CI +0.56 to +0.64), with controls within 0.01, while exact match fell from 0.07 to 0.004, answers grew from 2.9 to 16.5 words. Length adjustment removed about three quarters of the gain, and blinded raters (60 items per language, $\kappa = 0.38$ for support) judged the answers no more correct and, on average, less supported; the Malay change was +0.008. Together: an NLI reward can do nothing or do harm, and the score alone cannot tell which.

5.2.2 Justification

Zero-weight and shuffled-reward controls separate the reward's information from other effects. Reward-independent metrics, an independent scorer and a matched-length analysis are reported beside F_{faith}, because a metric that is also the reward cannot reveal its own divergence. The mT5 baseline shows what the reward does when it is active, so the null on the new backbones is not read as a harmless reward.

Limitations carried into Study 3. (i) Two small backbones, one reward weight ($\lambda_1 = 0.3$) and one NLI scorer family that is both reward and evaluator. (ii) The memorization explanation is a hypothesis; no run varies training length or logs reward variance. (iii)

Human validation exists only for the mT5 baseline. (iv) No mitigation was tested (a length penalty, a length cap or a question-conditioned reward). (v) Arabic (extractive) and Malay (multiple-choice-derived) differ in benchmark as well as language.

5.3 Objective O3 (Planned): Controlled Mitigation and Retrieval Robustness on Qwen3

5.3.1 Methodology (planned)

Study 3 is a staged design in which the mandatory stages change one factor at a time. Within-backbone objective comparisons are primary; scale and retrieval conditions are tested in later, separate stages.

Defining proxy divergence. Three questions are kept apart: does the automatic score agree with humans (*scorer validity*), does support alone make a good answer (*proxy sufficiency*), and does the score stay meaningful once it is optimized (*resistance to optimization*)? Divergence concerns the third. For each language–benchmark pair, the NLI-reward model is compared with SFT: ΔS is the change in the NLI score and ΔQ the change in the human-rated supported-and-correct rate. *Divergence* is concluded when the score clearly rises ($\Delta S \geq 0.10$, interval above 0.10) but no human-rated benefit as large as $+0.08$ is compatible with the data (divergence means a meaningful gain is excluded, not that quality fell; a smaller established gain is reported as such); a fall in quality is called *harm*, and rising quality is *agreement*. The five possible outcomes are mutually exclusive (Table 14, Appendix B), the margins are proposals fixed before any run, and the conclusion is falsifiable: a clear human-rated gain would reverse it. In Study 2, length is associated with the score change, and causation is not claimed; Study 3 tests it by intervention (the length-controlled arms) and reports the share of answer words copied from the passage, to separate verbosity from copying. Only the legacy mT5 Arabic result meets the score criterion (95% CI $+0.56$ to $+0.64$), and its human outcome is only suggestive; on Qwen3 and T5Gemma 2 the score movement was not established.

Stage 3a: reward-activity and pair-availability check. Study 2 found the NLI reward inert on Qwen3-0.6B-Base, probably because cross-entropy training memorizes the training set. Before any arm is compared, Stage 3a runs the SFT stage with early

stopping on the development set and reports, for each language, the within-prompt variance of reward, the share of steps with a non-zero policy-gradient term, and the share of training questions that yield at least one valid preference pair from eight samples. If a memorized SFT model yields too few pairs or no reward variance, an earlier checkpoint, chosen by a rule fixed before the runs, builds the candidate pool. *Gate:* both outcomes are admissible; if the NLI-reward arm stays inert, it is reported as such and the divergence test D is read from the preference arms. T5Gemma 2 (Zhang et al., 2025) is an optional second backbone if compute allows.

Stage 3b: controlled mitigation. On Qwen3-0.6B-Base, with the same data, split, seeds, decoding limits and evaluation, six arms are compared: (1) supervised cross-entropy fine-tuning (SFT); (2) the scalar NLI reward of Study 2; (3) ordinary DPO (Rafailov et al., 2023) on all valid pairs; (4) *volume-matched ordinary DPO*, a random subsample of the arm-(3) pairs of the same size as arm (5), matched on the distribution of scorer-A margin; (5) *length-controlled DPO*, the proposed method; and (6) a length-control baseline that receives the *same supervision* as the preference arms: a scalar reward of 1 for a supported-and-correct candidate (scorer A and the gold check) and 0 otherwise, with a length penalty on words above the development-set median of supported-and-correct answers; the penalty weight is tuned on the development set only, and no arm-specific hard cap is used, because the shared decoding limit applies to all arms. All arms use greedy decoding with the same maximum number of new tokens, so that no arm benefits from a looser length limit.

Preference-data construction. A *single common candidate pool* feeds every preference arm: for each training question, eight candidates are sampled from the SFT model at temperature 1.0. Each candidate is labelled supported (entailed by the passage according to scorer A, an NLI model used only for construction) and correct (token-F1 against the gold answer above a threshold fixed on the development set, or exact option match for Belebele). A pair takes a supported-and-correct candidate as *chosen* and an unsupported or incorrect candidate as *rejected*. For arm (3), all valid pairs are kept. For arm (5), the two answers must in addition have word counts within 20% of each other (the longer is at most 1.2 times the shorter, in whitespace-separated words, with token counts reported alongside). Pairs with ties, empty or truncated answers, or a

chosen answer that fails the gold check are removed from every arm. The number of pairs surviving each filter, and the mean length, scorer-A margin and question difficulty of the surviving pairs, are reported per arm, so that any difference in training volume or selection quality is visible. The 70 development questions of each pair, kept apart from the 399 (XQuAD-Arabic) or 401 (Belebele) training questions, are used to choose $\beta \in \{0.05, 0.1, 0.3\}$, the learning rate, the correctness threshold and the checkpoint; at about four pairs per question this gives roughly 1,600 pairs per language before length filtering, with arm (5) retaining a smaller fraction that is reported. If a pilot shows the volume to be insufficient, extra training-only data (TyDi QA Arabic, and Malay passages with generated questions) are added to all arms equally and kept strictly separate from every held-out set. A small set of unanswerable training items (gold passage removed) uses a fixed abstention phrase as the chosen answer. Scorer A is never used for evaluation, which relies on human ratings and a separately trained scorer B. The existing 716 Arabic and 430 Malay held-out examples are touched only for the final test. Each arm is run with five seeds.

Scorers. Two NLI scorers are fixed before any run: scorer A builds the training pairs and scorer B evaluates, so the model never trains on the scorer that judges it. Each answer is scored against each supplied passage separately (up to 512 tokens each) and the maximum entailment is used, so multi-passage input never exceeds the classifier window; the share of truncated inputs is reported. If a language fails validation, scorer-based construction is dropped for that pair: preference labels then come from the gold-answer check and human labels, scorer B stays evaluation-only, and human ratings remain the primary endpoint. Both are validated against human labels in each language before use (Appendix B, Table 15).

Novelty of arm (5). Length-controlled DPO changes the data, not the loss: it adapts known length-debiasing ideas (Park et al., 2024; Lu et al., 2024) to evidence-grounded training and is tested for added value, not claimed as a new objective. SamPO changes the loss and is not an arm; it is added as a descriptive comparison only if compute allows.

Limitation: no LLM judge. Neither training nor evaluation uses a large language model as judge. NLI is the object of study (the thesis tests whether such a score survives optimization) and is free, deterministic and reproducible as a reward, whereas an LLM

judge is costly, changes with the model version, is not validated for Arabic and Malay, and is known to favor longer answers, which is the bias under study; used as both reward and evaluator it would hide divergence. Evaluation therefore rests on blinded human ratings and an independent NLI scorer that is never trained on. DPO is chosen over the online reward because Study 2 found the online reward inert, and its preference pairs come from scorer A and the gold-answer check. An LLM judge could serve as a cheap secondary evaluator once validated against the human labels in each language; this is left to future work.

Supervision parity and attribution. Arms (3) to (6) use the same labels (scorer A support and the gold correctness check), the same candidate pool, the same SFT initialization, the same number of optimizer updates and the same hyperparameter budget on the same development set; arm (2) keeps the NLI-only reward of Study 2 because it replicates Study 2, so any difference between arm (2) and arms (3) to (6) is a *system comparison* (richer supervision and objective together) and is not attributed to the preference formulation. Table 9 states which contrast isolates which factor. The fraction of answers truncated by the shared decoding limit is reported per arm, so that no arm appears better because of incomplete answers.

Table 9: What each Study 3 contrast isolates.

Contrast	**Factor isolated**	**Residual confounds**
(2) vs (1) SFT	Effect of the NLI-only reward over supervised training (replication of Study 2)	Single scorer
(6) vs (1)	Effect of reward-based training with full supervision and length penalty	Penalty and reward bundled
(3) vs (6)	Preference formulation against scalar reward, same supervision; length not matched in (3)	Length differs
(5) vs (4)	Length filtering, at equal pair count and scorer-A margin	Question mix may differ
(5) vs (3)	Length filtering plus smaller, differently selected training set (not attributed to length alone)	Volume, selection
(5) vs (6)	Length-controlled preference training against scalar reward with a length penalty, same supervision	Optimization method

For (5) vs (4), balance diagnostics (retained-question overlap, question difficulty by scorer-A margin and by development-set accuracy, answer structure) are reported; where the retained questions differ, a question-matched resampling of arm (4) to the question set of arm (5) is added as a sensitivity analysis. A question that yields no valid pair among the eight candidates is reported (zero-pair rate and its difficulty), and SFT

keeps its full training set, so the preference arms see a subset of questions, which is stated as a limitation.

Data and splits. Belebele and XQuAD are documented as test or evaluation resources, so Study 3 uses them in an *adapted evaluation setting*: our own passage-disjoint split (XQuAD-Arabic: 399 training, 70 development and 721 test questions; Belebele: 401, 70 and 429; no passage appears in two partitions and all overlap checks are zero). Scores are therefore not comparable with published benchmark results. One model is trained per language–benchmark pair. The measured counts and checks are in Appendix B (Table 16).

Stage 3c: retrieval robustness. The trained generators are held fixed while the evidence condition varies: (i) oracle evidence only; (ii) oracle evidence plus $k = 2, 4, 8$ distractor passages (the top-ranked non-gold passages retrieved for the question, so they are topically similar; k counts distractors and not total passages), with the gold position randomized (first, middle, last); a truncation audit confirms that the gold passage is never cut, which separates position sensitivity from truncation; (iii) missing evidence, where the correct behavior is to abstain; and (iv) actual retrieval from a corpus. The corpus is described precisely as a *controlled Wikipedia-augmented benchmark*, not as an open-domain benchmark: a fixed, dated snapshot of the Arabic and Malay Wikipedia dumps, chunked into passages of about 100 words with 20% overlap, into which the gold passages of the benchmarks are injected. The retriever is the hybrid BM25 and dense system of Study 1, with BGE-M3 (Chen et al., 2024a) as the dense encoder and a cross-encoder reranker, passing the top 5 passages to the generator (sensitivity at 3 and 10).

Relevance and answerability. Injected gold identifiers are strict relevance labels; because they are incomplete, retrieval is also scored leniently (identifier match or scorer-B entailment of the gold answer). *Context-bound answerability* (the supplied passages support the gold answer) defines expected behavior in every condition and alone drives the abstention metrics; *corpus-wide answerability* only characterizes condition (iv). Condition (iii) removes the gold passage and expects abstention from the supplied passages. Supplied passages are screened by scorer B and an alias search, flagged items

are excluded and counted, and 100 unflagged items per language are checked by two annotators; if the false-negative rate exceeds 10%, condition-(iii) results are labelled exploratory. Abstention is scored semantically, with false abstention under condition (i) and coverage reported beside correct abstention. Abstention training is identical across arms: the same unanswerable items with the fixed phrase as target enter every arm as supervised targets, and the preference arms add them as pairs of the abstention phrase (chosen) against a sampled substantive answer (rejected), to which the length filter of arm (5) is not applied. Correct abstention is the fixed phrase or a paraphrase from a pattern list fixed before evaluation and confirmed by annotators in the human-rated block. The snapshot, chunking and retriever are frozen and hashed before generator evaluation, the retriever is tuned on development data only, and the share of injected passages nearly duplicating Wikipedia chunks is reported.

Stage 3d (optional): scale. If resources permit, the two strongest arms of Stage 3b are repeated on Qwen3-1.7B and Qwen3-8B (QLoRA (Dettmers et al., 2023) for the 8B model), so that scale varies while architecture and tokenizer are held fixed. Scale is never combined with a change of objective or evidence condition.

Harmonized language arm. Cross-language contrasts use only Belebele Arabic (arb_Arab) and Malay (zsm_Latn) with an identical prompt; XQuAD-Arabic is reported separately, and all other Arabic–Malay differences are differences between language–benchmark pairs.

Statistics in brief. (i) All comparisons are paired on the same test questions, grouped by passage, and reported with confidence intervals. (ii) Success for the main hypothesis needs a gain of at least $+0.08$ in human-rated supported-and-correct answers with the lower bound above zero, token-F1 not falling by more than 0.03 and answers not exceeding $1.5\times$ the SFT length, in a given pair. (iii) Because three language–benchmark pairs are tested, stricter confidence levels are used so that a win is not claimed by chance. (iv) The divergence test D compares the NLI-reward model with SFT (Table 14). (v) Robustness under distractors and abstention is reported descriptively. The full test inventory, resampling algorithm and multiplicity rules are in Appendix B (Table 17).

Table 10: Pre-specified decision criteria for Study 3. Margins are proposals, to be fixed before any run. H2d and H2e are secondary (descriptive intervals). The divergence test D (arm 2 vs arm 1) is classified by Table 14. The primary outcome is the human-rated supported-and-correct rate; token-F1 is the primary correctness metric and exact match (EM) is secondary.

Part	**Outcome and comparison**	**Success criterion**	**If not met**
H2a	Human supported-and-correct rate, length-controlled DPO vs SFT (oracle evidence)	Point estimate $\geq +0.08$ and Holm-adjusted lower 95% bound > 0 across the three language–benchmark pairs	No reliable improvement established (inconclusive or no gain)
H2b	Token-F1 (primary correctness metric), same contrast	Lower 95% bound of ΔF1 > -0.03	Non-inferiority not established; report whether estimates indicate harm (upper bound of $\Delta < 0$) or are inconclusive
H2b′	EM (secondary), same contrast	Lower 95% bound of ΔEM > -0.05	Reported; does not alone block H2a (F1 is the primary correctness metric)
H2c	Mean answer length, same contrast	At most $1.5\times$ the SFT mean	Reported as length inflation
H2d	Support and token-F1 under conditions (ii) and (iv), answerable items only	Non-inferior to SFT within 0.03 (lower 95% bound of the difference > -0.03)	Robustness not established; report per condition whether results indicate harm or are inconclusive (no claim that the benefit is limited to oracle evidence unless an interaction with evidence condition is tested)
H2e	Condition (iii), context-bound, screened items only: correct-abstention rate; and false-abstention rate under condition (i)	Correct abstention at least $+0.10$ above SFT, and false abstention at most $+0.03$ above SFT; gold-answer accuracy is *not* computed for these items	No abstention benefit claimed
D	NLI-reward arm vs SFT, human supported-and-correct rate and ΔS (oracle evidence)	Table 14: divergence if $L_S \geq 0.10$ and $0 \leq U_Q < +0.08$ (98.33% intervals)	Reported as harm, agreement or inconclusive
Control	Length-controlled DPO vs the length-control baseline, and vs volume-matched ordinary DPO, on H2a (both in the primary human-rated block; confirmatory)	Holm-adjusted lower 95% bound of the difference > 0	*No additional benefit over the control demonstrated*

Human evaluation. Each language has at least three qualified annotators (native or advanced proficiency, screened on a short test), compensated and trained in a calibration round of 40 items with written guidelines. Ratings are blinded to arm and to every automatic score, presented in random order, and a third annotator, who is not an author of the method, adjudicates disagreements. Weighted κ (and Krippendorff's α) must reach 0.6 on the calibration round before the main round starts; otherwise the guidelines are revised and calibration repeated. Ethics approval or written confirmation of exemption is

obtained before annotation starts.

Human evaluation sample. Humans rate 400 questions per language–benchmark pair for five arms (SFT, NLI reward, length-controlled DPO, length baseline, volume-matched DPO), each answer rated by two annotators, at one pre-selected seed per arm: about 15,700 ratings and about 197 annotator-hours (a planning figure). Four hundred questions can detect a difference of about $+0.08$; smaller effects are reported as inconclusive, and the divergence test D is powered for an absent or negative human-rated effect. No claim is made about human-rated consistency across seeds. Allocation, power and feasibility are in Appendix B (Table 18).

Annotation rubric. Three labels are recorded separately: *supported* (everything stated is in, or follows from, the passage), *correct* (it gives the gold answer) and *relevant* (it answers the question asked). An answer is supported-and-correct only if it is supported and correct; an extra true-but-unsupported claim breaks support, never correctness. The full rubric, a worked example and the abstention rule are in Appendix B.

Negative-result pathway. Study 3 admits any outcome of its first two stages. If Stage 3a finds the NLI-reward arm inert, as in Study 2, or finds no score–quality divergence on Qwen3, the thesis reports it, and Stage 3b is reframed as a test of whether any objective improves human-judged quality. If length-controlled DPO does not improve the primary outcome, or shows no additional benefit over the length-control baseline and the volume-matched control, the thesis result is the detection protocol and the evidence on where proxy divergence does and does not occur, and Study 3 is reported as a controlled negative result.

Minimum viable thesis and claim impact of each fallback. The mandatory core of O3 is Stage 3a, the primary human-rated block of Stage 3b (five arms) and the oracle, distractor and missing-evidence conditions of Stage 3c. Table 11 states which claim survives each fallback, in the order in which fallbacks would be applied.

If the mandatory core cannot be completed, O3 is narrowed with the supervisors before work continues.

Table 11: Fallbacks, in order, and the thesis claim that remains.

Dropped	**Claim that remains**	**Effect on O3, title and abstract**
Stage 3d (scale)	Within-backbone comparisons on Qwen3-0.6B-Base	None (optional)
Descriptive arms and evidence-condition human ratings	Primary comparisons rated by humans; robustness findings are automatic-only and labelled suggestive	Robustness stated as suggestive
Five seeds to three	Wider seed uncertainty for the automatic analysis	None
400 to 300 questions per pair	Larger detectable difference (about $+0.09$); margins re-fixed only before any run	H2a stated with the new margin
Condition (iv), actual retrieval	Conclusions limited to controlled evidence conditions (oracle, distractors, missing)	O3 and the aim are reworded to "controlled retrieval conditions"; the title keeps "controlled mitigation" and the abstract states the restriction

5.3.2 Justification (planned)

The stages isolate one factor at a time because changing model, scale, objective and retrieval setting together would make any improvement unattributable. Qwen3-0.6B-Base is the primary backbone because it is scale-matched to mT5-base, covers Arabic and Malay in pre-training, and reached Arabic exact match of 0.3–0.4 in the earlier legacy runs (exploratory figures from the superseded protocol, not a baseline; Study 2 and Stage 3a measure it), so the near-zero floor of mT5-base is not expected to be shared. DPO is compared, not assumed: it removes the sampled-reward estimator and λ_1 but keeps a trade-off parameter β and is itself length-biased (Park et al., 2024). Length control is therefore built into pair construction; established length-debiasing DPO variants such as SamPO (Lu et al., 2024) are not included as arms, to keep the factorial tractable, and are named as an untested alternative remedy. The length-control baseline tests whether it is the preference formulation or the length correction that matters. Preference labels come from scorer A and a gold check, while evaluation uses human ratings and scorer B, so the training signal is not the judging signal. Abstention under missing evidence is measured (Table 2) because a system that always answers cannot be faithful when evidence is insufficient.

Feasibility and fallback. Planning estimates, to be replaced by measured values from Stage 3a: Stage 3a about 100 GPU-hours (four arms, three language–benchmark pairs, five seeds: 60 models); Stage 3b about 270 (six arms, three pairs, five seeds: 90 models,

one model per pair, 0.6B backbone); Stage 3c about 75 plus indexing; optional Stage 3d about 100. These exclude the development search ($\beta \in \{0.05, 0.1, 0.3\}$ and two learning rates, one seed, about 20% of Stage 3b), candidate generation and the two NLI scorers, which are added after the pilot together with a 30% contingency. Human annotation is about 15,800 ratings (Table 18). Data licences are resource-specific: XQuAD, CC BY-SA 4.0; the Belebele dataset, CC-BY-SA 4.0 (its authors' assembled training set is CC-BY-NC and is not used); Wikipedia, CC BY-SA. The release contains split indices and the frozen manifest, prompts, derived annotations and model outputs under CC BY-SA 4.0, and the candidate's own code; the benchmark texts are not redistributed and users obtain them from the official sources. Adapter weights are released with a note that the licence status of weights trained on share-alike data is unsettled. Fallbacks follow Table 11, and each stage ends with a decision point in the Gantt chart (Table 12).

5.4 Data, Licensing, and Ethics

All experiments use publicly released benchmarks: XQuAD (Artetxe et al., 2020) for Arabic and Belebele (Bandarkar et al., 2024) for Malay, under their respective open licences, with a Wikipedia-augmented evaluation corpus to be specified for Study 3. XQuAD covers Modern Standard Arabic only, so the Arabic findings should not be generalized to dialectal Arabic. No personal data are processed. Study 2 used volunteer annotators (two per language) to rate model generations without access to the experimental condition, in an original round of 350 generations and a corrected round of 60 items per language; no demographic or personal information was collected, and the ethics status of the annotation tasks is declared in the Study 2 manuscript and is being confirmed with the supervisory team. Study 3 will use qualified, compensated, calibrated annotators under confirmed ethics approval or documented exemption, obtained before annotation starts. All code, adapter weights, and evaluation scripts will be released as described in Section 7.

6. Gantt Chart

Table 12 details the research activities, milestones, and publication plan across candidature, aligned to the candidate's actual enrollment and a targeted six-semester completion:

Studies 1–2 and this Proposal Defense fall within Semesters 1–2 (completed or in progress), Study 3 spans Semesters 2–5, and thesis compilation and the viva voce occupy Semesters 5–6.

Table 12: Research timeline by semester (six-semester completion target).

Activity	S1	S2	S3	S4	S5	S6
Study 1: Systematic review & pilot (Completed)						
PRISMA search, screening, corpus (61 studies)						
Failure-mode formalization & LoRA mapping						
Pilot experiments (Arabic retrieval, Malay baseline)						
Manuscript writing & submission (ACM TALLIP)						
Study 2: Controlled NLI-reward evaluation (Completed)						
First study: variant matrix (188 runs)						
Audit, controls, six-defect correction						
Primary study, 84 runs (Qwen3, T5Gemma 2) and mT5 baseline						
Manuscript writing, peer self-review, submission (JKSUCIS)						
Proposal Defense (this document)						
Study 3: Controlled mitigation and retrieval robustness (Planned)						
3a reward-activity and pair check on Qwen3-0.6B; ethics confirmation; update literature search						
3b preference-data construction and audit; six-arm comparison						
Human validation (calibration, main round); scorer B						
3c retrieval robustness and Wikipedia-augmented corpus; 3d optional scaling						
Multi-seed replication & statistical testing						
Manuscript writing & submission						
Thesis compilation						
Draft full thesis chapters						
Internal review, correction, viva preparation						
Thesis submission & viva voce						

Legend: completed activity planned activity milestone (this PD session)

Note: The timeline is compressed to meet a six-semester completion target. Corrected reruns (S2–S3), DPO training and human validation (S3–S4) overlap, and thesis drafting begins in S5 alongside Study 3 replication and writing. The overlap carries schedule risk, which is mitigated by reusing the Study 2 pipeline, evaluation code, and statistical protocol.

7. Dissemination

Completed means the work and manuscript are finished; *released* means code or data are public; *submitted*, *accepted* and *published* are the venue's stages. No study is yet accepted or published. The results are disseminated primarily as peer-reviewed journal publications, one per study, following the three-stage structure described in Section 3.3.

(a) **Study 1.** "Formalizing Retrieval-Augmented Generation Failure Modes and LoRA-Based Mitigation: A PRISMA-Guided Systematic Review with Pilot Closed-Passage Question Answering Experiments on Arabic and Malay." Target venue: *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)*. Status: manuscript complete; not submitted.

(b) **Study 2.** "An NLI Faithfulness Reward Added Nothing to LoRA Fine-Tuning of Qwen3 and T5Gemma 2 in Arabic and Malay Question Answering: A Controlled Null Result with a Length-Matched Analysis." Target venue: *Journal of King Saud University – Computer and Information Sciences (JKSUCIS)*, double-anonymized review. Status: manuscript complete; submission pending the ethics determination for the annotation tasks.

(c) **Study 3** (planned). Working title: "Length-Controlled Preference Optimization and Retrieval Robustness for Evidence-Grounded Answering in Arabic and Malay." Target venue: a leading NLP conference or journal, to be confirmed. Status: planned (Section 6).

In addition to publication, each study is accompanied by a public release: for Study 1, the review corpus and, once its component-level record (Supplementary S2) is complete, the pilot scripts, adapter weights and index configuration; for Study 2, the corrected training code, per-example predictions for all 84 runs, pinned dependencies and evaluation scripts; and for Study 3, the planned DPO pipeline and corpus configuration.

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A. Chronology of Study 2 development and runs

Dates are taken from the repository history (commit times, Malaysia time) and run logs.

Date (2026)	Event
17–22 Aug	Repository created; first-protocol mT5-base runs (LoRA, DoRA, QLoRA, AdaLoRA, VeRA; Arabic and Malay; Config B and C; seed 42, then 123 and 2026). Gradient checkpointing was applied to Config C only and to non-QLoRA variants only; held-out generation ran without `model.eval()`. 21 Aug: tokenizer fix (slow tokenizer for mT5 and XLM-R).
5–21 Sep	Qwen3-0.6B-Base and further mT5 runs under the first protocol (K=4 Malay rerun, 14–16 Sep); logging of all held-out answers and a λ_1 sweep added (14–19 Sep); human-annotation sample prepared (19 Sep).
7–8 Oct	Plain-LoRA baseline; human validation labels (350 generations, first protocol); gradient-checkpointing control flag; robustness analysis.
10–11 Oct	T5Gemma 2 runs (36 runs) and reanalysis, first protocol (superseded by the 84-run rerun).
12 Oct	Aligned REINFORCE scoring (sampler and scorer on the same distribution); empty-output audit, same-seed repeat, equivalence tests, estimator checks.
13 Oct, 14:40	Corrected-study commit: checkpointing audit, attribution controls, estimator fix.
13 Oct, 20:56	**Pre-specification commit** (v2): corrected training path and evaluation state; runbook fixing 8 seeds (42, 123, 2026, 7, 11, 13, 17, 19), the primary estimand (pooled C−B change in F_{faith}, seed-paired), equivalence margin ± 0.01, zero-weight and shuffled-reward controls, and the analysis script, before any v2 result existed.
13–14 Oct	v2 GPU runs (core 48 jobs, controls 96, checkpointing factorial 18; 162 jobs, 228 records, 0 failures); independent-scorer pass (14 Oct, 04:31). The policy-mode, AdaLoRA/VeRA and frozen/full fine-tuning stages of the runbook were not run.
14 Oct	Analysis and human-validation scoring; run archive committed (11:44); repository restructured (20:28).
18–19 Oct	**Primary study** (hero_v2): Qwen3-0.6B-Base and T5Gemma 2, 5 seeds, 84 runs: 72 of 84 runs done 19 Oct 03:42, all 84 at 09:33; analyses committed 19 Oct.

Deviations and notes. (i) The runbook planned 100 items per language for the human check; 60 per language were analysed. (ii) The first-protocol results informed the design of the corrected study (for example the ± 0.01 margin was set from the first study's pooled effect of 0.018), so the mT5 study is a corrected re-run, not a fresh test. (iii) Run logs carry no date; the 13 October launch is inferred from log time and commit order.

B. Technical specification (supporting detail)

The main text gives key takeaways; this appendix keeps the full specifications that an examiner may wish to check.

Study 2 estimator and training settings

Study 2 extended the Stage 2 pilot of Study 1 into a controlled closed-passage evaluation of a composite objective $L_G = L_{CE} + \lambda_1(1 - F_{\text{faith}})$, with $\lambda_1 = 0.3$, in which F_{faith} is an NLI-entailment score of the model's own samples, optimized with a leave-one-out REINFORCE estimator ($K = 4$ samples). As implemented, the gradient is that of $L_{CE} + \lambda_1 \left(-\frac{1}{BK}\sum_{i,k} A_{ik} \log \pi_\theta(y_{ik} \mid x_i)\right)$, where y_{ik} are sampled answers (temperature

1.0, full softmax, at most 40 new tokens), $A_{ik} = F_{\text{faith}}(y_{ik}) - \frac{1}{K-1}\sum_{j\neq k} F_{\text{faith}}(y_{ij})$ is a detached leave-one-out advantage, $\log \pi_\theta$ is the sum of token log-probabilities (not length-normalized), and the frozen NLI model supplies a scalar reward only, with no gradient through it; L_G above is the objective this estimator targets, not a differentiable loss. Sampling and re-scoring run with dropout off. The primary study ran on Qwen3-0.6B-Base and T5Gemma 2 270M-270M (470 training examples; 720 Arabic and 430 Malay held-out examples; five seeds; 84 runs), and the legacy baseline on mT5-base (580M; 716 Arabic and 430 Malay examples after four overlapping Arabic items were excluded), with rank 8, $\alpha = 16$ and AdamW at 2×10^{-4}.

Formal definition of proxy divergence

Three properties of a proxy score S (here the NLI support score of scorer A, the reward scorer) are distinguished, each with its own estimand. (a) *Scorer validity*: agreement of S with blinded human support labels on a held-out set (rank agreement, area under the curve, calibration), per language. (b) *Proxy sufficiency*: whether support alone suffices for answering, estimated as the association of S with the human supported-and-correct rate and with token-F1 at fixed answer length. (c) *Resistance to optimization*, which defines divergence. For each language–benchmark pair, let ΔS be the change in S produced by the NLI-reward arm (2) relative to SFT (arm 1), and ΔQ the change in the primary outcome Q (human supported-and-correct rate; token-F1 secondary) between the same two arms. The margins $\delta_S = 0.10$ on the 0–1 NLI score and $\delta_Q = +0.08$ (the improvement that H2a requires) are proposals, fixed before any confirmatory run. Both changes carry two-sided 98.33% intervals ($1 - 0.05/3$, Bonferroni over the three pairs), $[L_S, U_S]$ and $[L_Q, U_Q]$. ΔS is estimated on the five automatic seeds by a hierarchical bootstrap over seeds and then passages, and ΔQ on the double-rated primary block (400 questions per pair, one selected seed per arm; test D of Table 17) by a passage-clustered paired bootstrap. Table 14 assigns every outcome of a pair to *exactly one* category. For example, $\Delta Q = +0.06$ with interval $[+0.01, +0.11]$ and $L_S \geq \delta_S$ is *inconclusive* (rule 5), because $U_Q \geq \delta_Q$ and the point estimate is below δ_Q. A divergence conclusion is falsifiable: an annotation that yields $\Delta Q \geq \delta_Q$ with $L_Q > 0$ would reverse it. On this definition the Study 2 Arabic result meets the ΔS criterion (95% CI $+0.56$ to $+0.64$), whereas its ΔQ classification is *suggestive* (small, low-agreement human sample) and is reported as such. Failure of the scorer itself (a), failure of treating support as sufficient (b), and divergence under optimization (c) are different findings and are reported separately.

Table 14: Mutually exclusive and exhaustive classification of a language–benchmark pair (arm 2 vs arm 1). Rules are disjoint by construction; the order is only a reading aid.

Rule	Category	Condition	Reading
1	Score movement not established	$L_S < \delta_S$	Divergence not tested for this pair
2	Harm	$L_S \geq \delta_S$ and $U_Q < 0$	Score up, quality demonstrably down
3	Divergence	$L_S \geq \delta_S$ and $0 \leq U_Q < \delta_Q$	No benefit as large as δ_Q is compatible with the data (a smaller established gain is reported as such)
4	Agreement	$L_S \geq \delta_S$, $U_Q \geq \delta_Q$, $L_Q > 0$ and $\Delta Q \geq \delta_Q$	Score and quality rise together
5	Inconclusive	$L_S \geq \delta_S$, $U_Q \geq \delta_Q$ and rule 4 not met	Data compatible with benefit and with none

Scorer specification

Table 15 fixes both NLI scorers before any run; both are used as black-box classifiers and neither is a substitute for human rating.

Human labels map to the binary target as supported = rubric label (1); neutral and contradictory both count as unsupported. Scorers A and B were trained on overlapping NLI data, so their independence

Table 15: Scorer specification (frozen and hashed before the first Study 3 run).

Item	Scorer A (construction, reward)	Scorer B (evaluation)
Checkpoint	`MoritzLaurer/mDeBERTa-v3-base-mnli-xnli`, revision pinned	`joeddav/xlm-roberta-large-xnli`, revision pinned
Input	Premise: each supplied passage in turn, with the maximum taken over passages. Hypothesis: the answer as generated (question not appended)	Same
Score	Softmax probability of the entailment label of the three-way output	Same
Truncation	Each passage truncated from the end to fit 512 tokens with the answer kept whole; passages whose gold evidence lies beyond the cut are flagged and counted, and results are reported with and without them	Same
Threshold	τ_A for "supported" (default 0.5), tuned per pair on the calibration and pilot annotations only	τ_B, tuned the same way, never on test items
Language validation	Neither model saw Malay NLI training data; per pair, AUC against human support labels (acceptance: AUC ≥ 0.70 with bootstrap lower bound > 0.60), calibration (Brier score and reliability bins) and κ at τ (≥ 0.4) on about 140 double-rated items per language	Same
Failure rule	If a pair fails validation, scorer-based claims for it are descriptive, and construction uses the gold-answer check and human labels only	Evaluation only; never used for construction

is partial; the correlation of their errors against human labels is reported. Before any training, a scorer test set of at least 20 hand-written items per language (answers copied from the passage, entailed but irrelevant answers, partly supported answers with one unsupported claim, contradictory answers, and the abstention phrase) is scored, and a scorer that rates a copied passage above the correct short answer is flagged as favoring length.

Splits, lineage and the adapted evaluation setting

Belebele is documented as a test-only benchmark and XQuAD as an evaluation corpus. Study 3 therefore uses them in an *adapted evaluation setting*: a controlled, passage-disjoint training, development and test resplit created for this experiment. It is not the standard protocol of either benchmark, scores are not comparable with published Belebele or XQuAD results, no state-of-the-art claim is made, and the CC-BY-NC training set assembled by the Belebele authors is not used. Partitions are generated once (script `make_split.py`, seed 2026) by *passage*: questions sharing a paragraph (XQuAD) or a passage (Belebele) go to the same partition, passages are also matched by normalized hash, and the Belebele grouping is identical in Arabic and Malay, so a parallel item never crosses partitions between languages. Overlaps of passage hash between partitions, of question text between partitions and of XQuAD with Belebele passages are all zero; the manifest (SHA-256 `db754155fa3a88b2dcd40b7caa19082b830bb384b1c7388cf00c001c95b03a31`) is frozen before any run. The grouped split (Table 16) is authoritative and replaces the contiguous split of Study 2 (470 training, 716 Arabic and 430 Malay held-out items): because it re-partitions all 1,190 XQuAD items, some Study 2 test items now lie in Study 3 training or development data, and any item inspected during Study 2 that falls into a Study 3 test set is an adaptive reuse, reported as a limitation. One model is trained per language–benchmark pair, from the base

checkpoint, on its own training questions only (399 or 401); each test set is evaluated once per arm after choices are frozen, and claims are limited to these benchmarks.

Table 16: Passage-disjoint split (measured, seed 2026): questions (passages).

Benchmark	Train	Development	Test	Total
XQuAD-Arabic	399 (83)	70 (16)	721 (141)	1,190 (240)
Belebele (Arabic and Malay, same indices)	401 (219)	70 (37)	429 (232)	900 (488)

Test families, resampling and multiplicity

The unit of inference is the question, clustered by paragraph or passage and paired across arms. *Resampling algorithm:* each replicate draws passages with replacement (all questions of a drawn passage move together), recomputes every arm's statistic on the same drawn questions (paired), and uses 10,000 replicates and percentile intervals; the five-seed automatic analysis first draws seeds with replacement and then passages. Intervals for a family of m tests are taken at level $1 - 0.05/m$ (Bonferroni) or by Holm adjustment, as stated in Table 17, which lists every test and its family. The confirmatory families are A, B and C; a general claim for a family needs all three pairs, and "improves quality" additionally requires H2b and H2c in the same pair (conjunctive). H2d and H2e are *secondary*: they are reported with 95% intervals and decision labels, without a multiplicity-adjusted confirmatory claim. The $+0.08$ margin is partly set by the planned sample and not derived from a user-need analysis; it is re-examined with annotators and supervisors before any run, a ± 0.02 sensitivity analysis is reported, and it is fixed before treatment effects are inspected. The evidence-condition blocks (60 questions) are descriptive; non-inferiority intervals under conditions (ii) and (iv) support only the statement that no reliable degradation was detected.

Table 17: Test inventory and families (per language–benchmark pair unless stated).

Test	Comparison (outcome)	Family, adjustment	Sample
A (H2a)	Arm 5 vs arm 1 (human supported-and-correct)	Confirmatory; Holm over 3 pairs	Primary block, 400 per pair
B (H2b, H2c)	Arm 5 vs arm 1 (token-F1 non-inferiority; length ratio)	Conjunctive with A in the same pair; no extra adjustment	Same
D	Arm 2 vs arm 1 (human supported-and-correct) with ΔS (Table 14)	Confirmatory; Bonferroni 98.33% intervals over 3 pairs	Primary block, 400 per pair; ΔS on the same selected-seed outputs (five-seed change reported separately)
C	Arm 5 vs arm 6 and vs arm 4 (human supported-and-correct)	Confirmatory only where A is established; Holm over at most 6 tests	Primary block
H2d, H2e	Conditions (ii), (iii), (iv) and false abstention	Secondary; 95% intervals, descriptive	60 per condition

Human-evaluation sample, allocation, power and feasibility

Human ratings cover one pre-designated seed per arm (the seed whose development-set rate of supported-and-correct answers, judged by scorer B and the gold check, is the median of the five seeds, with token-F1 breaking ties; computed once, before any test evaluation); the remaining four seeds are scored automatically (gold check and scorer B) and used to estimate seed variance and to check that the human-rated seed is not atypical. The human comparison therefore estimates the difference between the *selected* (development-median) checkpoints, which is the quantity a practitioner using a development-based selection policy

would obtain, and not human-validated consistency across training seeds. Presentation to annotators is randomized, arms of one question are shown in random order, support is judged and locked before the gold answer is shown, and the arm label is hidden; because answer length and style can still reveal the arm, length-balanced items are reported as a sub-analysis, and agreement on the main round is monitored and compared with the calibration agreement (a fall of more than 0.10 triggers re-calibration), with adjudication independent of arm identity. The same randomly sampled test questions are used for every arm, so that comparisons are paired. A language–benchmark pair is Arabic–XQuAD, Arabic–Belebele or Malay–Belebele. Table 18 gives the allocation.

Table 18: Human-rating allocation (outputs rated; each primary and condition item is rated by two annotators). Counts assume three language–benchmark pairs.

Block	**Arms**	**Questions per pair**	**Seeds**	**Outputs**	**Ratings**
Primary (oracle evidence): SFT, NLI reward, length-controlled DPO, length baseline, volume-matched ordinary DPO	5	400	1	6,000	12,000
Other arm (ordinary DPO), descriptive	1	150 (subset of the 400)	1	450	450 (single) + 20% audit
Evidence conditions (ii), (iii), (iv)	3	60 per condition	1	1,620	3,240
Total				**8,070**	**about 15,700**

Power. For a paired binary outcome with about 30% discordant pairs, 400 questions give a standard error of about 0.027 and 80% power (two-sided $\alpha = 0.05$) for a difference of about $+0.08$, which is why the H2a margin is $+0.08$ and not $+0.05$; with 150 questions the detectable difference is about $+0.13$, so the 150-item block is descriptive only. This is *preliminary*: the decision rule of Table 10 also needs a point estimate of at least $+0.08$ and Holm adjustment over three pairs (critical $z \approx 2.39$), and under the full rule the probability of success is about 50% if the true gain is exactly $+0.08$, 77% at $+0.10$ and 93% at $+0.12$. A simulation on the calibration discordance replaces it. Questions of one passage are correlated: with about 2.8 sampled questions per passage, an intraclass correlation of 0.1 to 0.3 raises the standard error by 9 to 25% (to about 0.030 to 0.034) and lowers the probability of concluding divergence at a true effect of zero from 70% to 62% and 48%; the simulation therefore resamples passages and is run before the margins are frozen. Test D's score change is computed on the same selected-seed outputs that humans rate. Smaller gains are reported as inconclusive. For test D the 98.33% bound U_Q lies about 0.065 above the estimate, so with $\delta_Q = +0.08$ divergence is concluded with probability about 70% if the true ΔQ is 0, 95% if -0.03 and 30% if $+0.03$; with $\delta_Q = +0.05$ the first figure would fall to about 30%, which is why the margin equals the H2a margin. Test D is thus powered to detect an absent or negative human-rated effect, not a small positive one, and an effect between 0 and $+0.08$ is reported as inconclusive; ΔS is large relative to its interval (legacy mT5 baseline: $+0.60$) and is not limiting. The primary block, including the volume-matched control and the NLI-reward arm, is mandatory and confirmatory (Table 18); descriptive arms and evidence-condition ratings are reduced first if annotator capacity falls short. Because human ratings cover one selected seed per arm, no claim is made about human-rated consistency across seeds; seed variance is assessed with automatic measures only.

Why the human evaluation is feasible. Answers are short (2.9 to 16.5 words in Study 2) and the passage is read once per question and shared by all arms, so the five primary arms cost about 1.7 times a single rating. At 45 seconds per rating, the 15,690 ratings of Table 18 plus about 90 audit ratings (15,780 in all) take about 197 annotator-hours before calibration and adjudication: roughly 44 hours each for three Arabic annotators and 22 hours each for three Malay annotators, over eight to ten weeks. The 40-item calibration round and the $\kappa \geq 0.6$ gate limit rework, only the primary block is confirmatory, scorer B

and the gold check cover all outputs, and a 100-output pilot in Stage 3a measures time per rating and discordance (above 90 seconds, the fallbacks of Table 11 apply). These are planning figures; calibration and adjudication hours are added to the measured budget.

Rationale for the divergence margin $\delta_Q = +0.08$

The margin δ_Q is the human-rated benefit below which a rise in the NLI score is called divergence. It is set equal to the H2a success margin, so that "no benefit as large as the one we would call success" is the same statement on both sides of the test. Table 19 shows the probability of concluding divergence for 400 questions per pair (standard error 0.027, critical $z \approx 2.39$). A smaller margin ($+0.05$) rarely concludes divergence even when the true effect is zero (29%); a larger one ($+0.10$) concludes it easily but would label a real $+0.09$ benefit as divergence. The margin is a proposal to be ratified by the supervisors at the proposal defense and is frozen, with the split manifest, before any confirmatory run; a later change is recorded as a dated amendment.

Table 19: Probability of concluding divergence (rule 3) by margin and true human-rated effect, 400 questions per pair, 98.33% intervals.

Margin δ_Q	**True $\Delta Q = -0.03$**	**True $\Delta Q = 0$**	**True $\Delta Q = +0.03$**
$+0.05$	0.70	0.29	0.05
$+0.08$ (proposed)	0.95	0.70	0.29
$+0.10$	0.99	0.90	0.57

Annotation rubric and worked example

Three labels are recorded separately and in this order. (1) *Supported*: every factual claim in the response is stated in, or directly follows from, the supplied passage(s), judged without seeing the gold answer. (2) *Correct*: the response gives, as its answer to the question asked, the same information as the gold answer (a paraphrase, or a superset that contains the gold answer without contradicting it, is correct); correctness concerns the requested answer only, so an extra claim that is true or unsupported does not make a response incorrect, but any statement that contradicts the gold answer does (this takes precedence). (3) *Relevant*: the response answers the question asked and not an entailed but different question. A response is *supported-and-correct* only if it is supported and correct. Extra claims therefore affect the support label and never the correctness label; an extra claim that is true in the world but absent from the passage is unsupported. For missing-evidence items the correct response is the fixed abstention phrase; any substantive answer is unsupported and, because it does not abstain, incorrect for that item.

Worked example (illustrative). Passage: "The bridge opened in 1932 and was designed by Ahmad bin Ali." Question: "Who designed the bridge?" Gold: "Ahmad bin Ali". "Ahmad bin Ali designed the bridge" is supported-and-correct. "Ahmad bin Ali designed the bridge, which is the longest in the country" is correct but not supported (the superlative is absent from the passage). "The bridge opened in 1932" is supported but incorrect and irrelevant. "Ali bin Ahmad" is neither supported nor correct.

C. Literature evidence record

This appendix keeps the comparison with the closest prior work and the update-search record (evidence matrix: Supplementary S1).

The systematic search of Study 1 closed in April 2025, whereas this proposal is dated November 2026. A targeted update search covering May 2025 to 17 October 2026 was run for this proposal: four queries (length bias in preference optimization for RAG; Arabic and Malay RAG fine-tuning; NLI rewards and proxy validity; distractors, missing evidence and abstention) returned 37 records, screened against the Study 1 criteria; eight were retained (Table 20). It is a *targeted* update, not a PRISMA replication; the protocol will be re-run on the institutional-access databases Study 1 used (Scopus, IEEE Xplore, ACL

Anthology), with Web of Science added, before the thesis is submitted. On 17 October 2026 the four queries were re-run in the same engine, adding SamPO (Lu et al., 2024) (a length-debiasing DPO remedy), Cai et al. (Cai et al., 2026) and an evidence-sufficiency detector (Chen et al., 2026). A second re-run on 18 October (Consensus index) retained six abstract-level works (Singhal et al., 2023; Chen et al., 2024b; Skalse et al., 2022; Li et al., 2025; Wang et al., 2026; Zhao et al., 2026) on length exploitation, formal reward hacking, length-controlled DPO and English grounding rewards. None changes the gap: length exploitation is known in general alignment, and the English grounding-reward works have no Arabic or Malay evaluation, length-matched baseline or human check. The audit status is: the evidence matrix (Supplementary S1) records twenty works, eleven read in full text and nine (ASC (Pham et al., 2026), GRAB-RAG, Cai et al. and the six above) at abstract level only; no gap statement rests on an abstract-level work alone. The institutional-access databases were not searched.

Three lines of earlier work are closest to Study 3. RAFT (Zhang et al., 2024) fine-tunes a model to answer from retrieved documents while ignoring distractors, which overlaps with Study 3's robustness stage (FM2) but is English, domain-specific and supervised, not preference-based. Retrieval Preference Optimization (Yan et al., 2025) injects the relevance of retrieved documents into preference optimization for robust RAG; it addresses retrieval-conditioned preferences, whereas Study 3 builds preferences from support and correctness with explicit length control and tests them against independent outcomes. Work on multilingual RAG (Park and Lee, 2025) shows that systems favor query-language and high-resource-language evidence, which motivates testing Arabic and Malay under actual retrieval but ignores training-time signals. Length is a known exploitable bias in preference optimization (Park et al., 2024), and reward over-optimization is documented for learned rewards (Gao et al., 2023a).

The retained records, discussed here and tabulated in Table 20, narrow the claim. First, the divergence between a checker-based reward and answer quality is not new in the abstract: Ji et al. (2026) diagnose signal collapse and reward hacking (very short answers, avoidance of search) for checker-guided RAG in biomedical English, and Xu et al. (2025) and Pham et al. (2026) propose finer-grained faithfulness rewards; the former validates its reward model on 100 author-annotated examples, and none of the three uses length-controlled preference baselines. Ji et al. (2026) add a graded length penalty after observing short-answer hacking, but have no shuffled-reward control and no human ratings. Study 2 is therefore positioned as a controlled, audited test in two low-resource languages (with shuffled-reward controls, not found in the works above), not as the first report of reward hacking. Second, distractor-resilience training (Kachuee et al., 2025), faithfulness-aware tuning (Huang et al., 2025) and abstention under missing or misleading context (Setiawan, 2026) (English, prompt-based) overlap with the robustness and abstention parts of Study 3; none tests Arabic or Malay or trains abstention with a length control. Third, XRAG (Liu et al., 2025) and Alsubhi et al. (2025) cover Arabic RAG evaluation and pipelines, with no Malay and no faithfulness training.

The gap is narrower than "RAG needs preference optimization" or "non-English evaluation is missing". As far as the April 2025 search and the targeted update (to 17 October 2026) show, no identified study (i) measures evidence support, correctness, length and abstention as separate constructs for an entailment-based training signal, (ii) tests a length-controlled preference construction against ordinary DPO, a volume-matched DPO and a simple length baseline on the same backbone and data, and (iii) does so for Arabic and Malay under missing, distracting and retrieved evidence with human validation.

Status. Targeted search only; the database re-run is planned. Cells were checked against full text on 17 October 2026 except those marked "abstract level only" (Supplementary S1); "Not stated" means only that the work does not report the item.

Table 20: Comparison of Study 3 with the closest identified work (April 2025 search and targeted update to 17 October 2026). Cells were checked against full texts on 17 October 2026, except where marked "abstract level only".

Work	Training signal	Evidence conditions	Languages	Length control; human rating
RAFT (Zhang et al., 2024)	Supervised (SFT), with distractors	Oracle plus distractors; some examples without the oracle (Sec. 3)	Not stated	No length control; no human rating

Work	Training signal	Evidence conditions	Languages	Length control; human rating
RPO (Yan et al., 2025)	Preference (accuracy-labelled pairs) with a retrieval reward	Relevant context; incorrect context in one test (Table 3)	Not stated	Length-normalized reward (Sec. 5.1), no length-matched pairs; no human rating
PrismRAG (Kachuee et al., 2025)	Supervised, distractor-aware	Gold plus word-count-matched distractors; abstention scored	English benchmarks	No length control; LLM judge, no human rating
RHIO (Huang et al., 2025)	Faithfulness-aware tuning with control codes (no DPO)	Sufficient evidence only	Not stated (English data)	No length control; 200 human ratings (2 annotators)
VERITAS (Xu et al., 2025)	RL (PPO), faithfulness reward	Search-based, Wikipedia	Not stated	No length control; 100 author-annotated examples (reward-model check)
Checker-guided RAG (Ji et al., 2026)	RL (GRPO), NLI-checker reward; zero-weight control present, shuffled-reward control not found	Biomedical QA, top-5 passages	English-focused	Graded length penalty added after short-answer hacking; no human rating
ASC (Pham et al., 2026)	RL, sentence-level credit	Long-form RAG	Not stated	Abstract level only
GRAB-RAG (Setiawan, 2026)	None (prompted abstention, frozen models)	Supportive, degraded, missing, misleading	NQ and HotpotQA (English)	Abstract level only
XRAG (Liu et al., 2025)	None (benchmark)	Gold, distractors, no retrieval	English, German, Spanish, Chinese, Arabic (no Malay)	No length control; human check of items and judge
DPO length margin (Park et al., 2024)	Preference, length-margin regularizer	None (summarization, dialogue)	Not stated	Regularizer, not matched pairs; GPT-4 judge
SamPO (Lu et al., 2024)	Preference, down-sampled KL	None (reasoning, summarization)	English	Token down-sampling, not matched pairs; 30-person binary human evaluation
Study 3	Reward, ordinary, volume-matched and length-controlled preference, length baseline	Oracle, distractors, missing, Wikipedia-augmented	Arabic, Malay	Explicit control; 400-item paired human rating